TEXORA · LECTURE SHEET

Migration at 80°C during Dyeing in Exhaust Method

Why a hot, pre-fixation hold levels a dyeing: desorption, transport and re-adsorption, the exhaustion–levelness trade-off, and how the migration index is measured.

TE-3105 · TEXTILE CHEMISTRY & COLORATION-II · TOPIC 7 · LEVEL 3 TERM 1

1. Migration — the levelling mechanism

Migration is the movement of dye already on the fibre back into the bath and onto other fibre regions during dyeing. It is the mechanism of levelling: dye that struck heavily on the first-contacted surfaces desorbs, travels, and re-adsorbs where the fibre is still pale. A dyeing can therefore start unlevel and finish level — if conditions keep the dye mobile long enough. Migration depends on dye substantivity (low-substantivity dyes migrate best), temperature, time, and the presence of levelling agents; it falls as fixation or strong binding locks dye in place.

2. What happens at 80 °C in exhaust dyeing

The syllabus fixes attention on 80 °C because it is the diagnostic temperature of reactive and direct dyeing practice: high enough for rapid diffusion in cellulose, below the boil, and the holding temperature of hot-brand reactive dyeing. In an exhaust bath raised to 80 °C and held there *before* alkali addition (reactive) or with salt present (direct):

- Diffusion coefficients in swollen cellulose roughly double for each 10–15 °C rise, so at 80 °C dye penetrates yarn interiors that it only coated at 30 °C.
- Adsorption equilibrium shifts towards the bath (adsorption is exothermic): exhaustion at equilibrium is *lower* at 80 °C than at 30 °C, but the dye on the fibre is far more evenly distributed.
- Migration index — measured by dyeing a pale companion fabric in the used bath conditions, or by the contact-migration test — peaks in this hot, pre-fixation window.

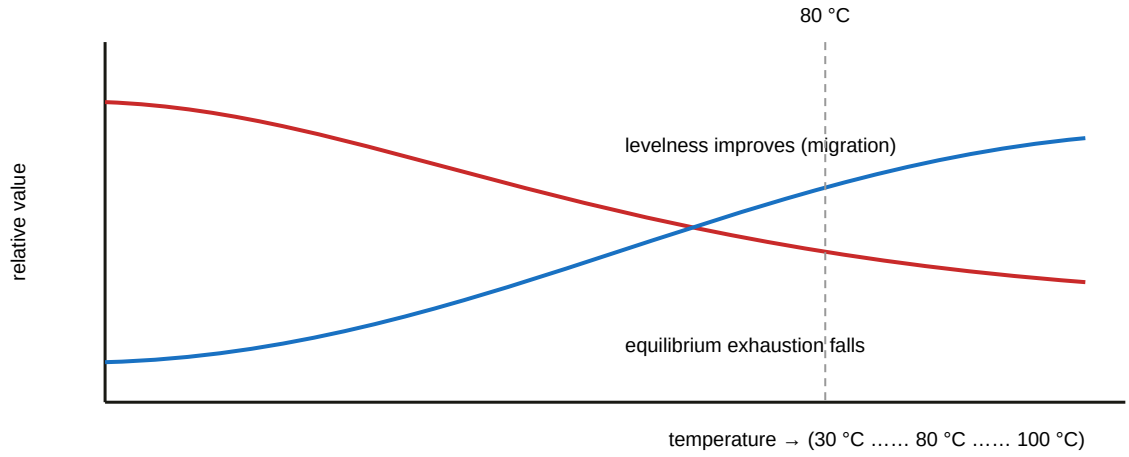


Fig. 1 — The migration trade-off: heating an exhaust bath towards 80 °C sacrifices some equilibrium exhaustion but buys levelness through migration.

3. The practical sequence the curve dictates

1. **Exhaust cool, level hot, then fix:** apply reactive dye at 30–40 °C with salt (high exhaustion, dye loosely held), raise to 60–80 °C by dye class and hold 15–30 min for migration/levelling, then dose alkali — fixation freezes the level distribution achieved.
2. **Direct dyes:** raise slowly to the boil with salt added in portions; the hot hold is where skittery (uneven) dyeings are rescued by migration, assisted by levelling agents (e.g. anionic/non-ionic surfactant blends) that moderate re-adsorption.
3. **Disperse dyes on polyester** migrate too, but only near 130 °C where chain mobility allows it; a levelness hold at top temperature serves the same purpose, and *thermomigration* during later heat-setting (dye moving to the fibre surface) is the harmful namesake to control.

4. Measuring migration

The standard laboratory check: dye fabric A to depth; then run A together with equal white fabric B in a blank bath (all chemicals, no dye) under the dyeing conditions. The depth B gains, as a percentage of A's original depth, is the **migration index**. Levelling-class direct dyes may transfer 40–60%; high-substantivity dyes under 15%. In reactive dyeing the index is measured before alkali addition — after fixation, migration of fixed dye is zero by definition, which is exactly why timing matters.

WORKED EXAMPLE — MIGRATION INDEX

Fabric A dyed to K/S = 12.0 at its λmax. After the blank-bath treatment with white fabric B: A falls to K/S 8.4, B rises to K/S 3.6 (K/S is proportional to dye concentration at moderate depths). Dye transferred = 3.6/12.0 = 30% of the original dye moved. Migration index = **30%** — a moderately migrating dye; a 20–30 min hot hold will level most application unevenness, but a

> badly struck start still shows. Mass balance check: 8.4 + 3.6 = 12.0 ✓ (no dye lost to the blank bath beyond the two fabrics).

5. Migration elsewhere in coloration

The same physics appears in pigment padding (migration during *drying* — Topic 4, unwanted), in printing (dye migration during steaming evens prints), and in storage (sublimation/thermomigration of disperse dyes). The exhaust-dyeing lesson generalises: wherever dye is mobile, gradients decay towards levelness; wherever it is locked, gradients are permanent. Process design is choosing when to be in each state.

6. Glossary — key terms

- **Migration index:** % of dye transferred to a white companion in a blank bath under process conditions.
- **Strike:** the initial rapid uptake of dye on first contact.
- **Skitteriness:** fibre-to-fibre unevenness from variable dye affinity.
- **Thermomigration:** movement of disperse dye to the polyester surface during dry heat.

7. Key points

- Migration = desorption + transport + re-adsorption; it is how levelness is earned.
- At 80 °C cellulose dyeings trade a little exhaustion for a lot of levelness — exploit the window before fixation.
- Fixed dye does not migrate: sequence exhaustion → hot levelling hold → fixation.

8. Summary

Migration at 80 °C is the exhaust method's self-correction mechanism, strongest exactly where hot-brand reactive and direct dyeings are held before fixation. Understanding its temperature and substantivity dependence turns unlevelness from a mystery into a scheduling problem: keep the dye mobile until the goods are level, then lock it.

9. Practice

1. In the worked example, suppose B had reached K/S 5.4 and A fallen to 6.6. Calculate the migration index and state whether a 15-minute hold would level a moderately uneven start.
2. Why must the migration hold in reactive dyeing come before alkali dosing, not after?

3. A direct dyeing is unlevel at 60 °C. The dyer adds more salt. Criticise the decision using migration principles.

Answer key

1. Index = 5.4/12.0 = 45% — a freely migrating dye; a 15-minute hot hold will substantially level moderate unevenness (check: 6.6 + 5.4 = 12.0 ✓).
2. Alkali converts adsorbed dye into covalently fixed dye, and fixed dye cannot desorb or migrate. Unevenness present at dosing is fixed permanently; the level distribution must be achieved first.
3. Salt raises substantivity/exhaustion and suppresses desorption — it locks the existing uneven distribution more firmly and speeds further uneven uptake. The correct moves are heat (towards 80–98 °C), time, and levelling agent to promote migration.

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